

DELIVERABLE 1

GROUP 7: GAME MASTER

GROUP MEMBERS:

1. DOSHI, PRATHAM MILANKUMAR
2. LAKHANI, VIMAL SHANTIBHAI
3. SHAHZAD, AZAAN
4. SOLANKI, NEEL PARESH

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Team Contract

SYST 17796 TEAM PROJECT

Team Name: GAME MASTER

Please negotiate, sign, scan and include as the first page in your Deliverable 1.

Please note that if cheating is discovered in a group assignment each member will be charged with a cheating offense regardless of their involvement in the offense. Each member will receive the appropriate sanction based on their individual academic integrity history.

Please ensure that you understand the importance of academic honesty. Each member of the group is responsible to ensure the academic integrity of all of the submitted work, not just their own part. Placing your name on a submission indicates that you take responsibility for its content.

For further information, read Academic Integrity Policy here :

<https://caps.sheridancollege.ca/student-guide/academic-policies-and-procedures.aspx>

Team Member Names (Please Print)	Signatures	Student ID
Project Leader: PRATHAM DOSHI	PD	991759009
 VIMAL LAKHANI	VL	991745909
 NEEL SOLANKI	NS	991754568
 AZAAN SHAHZAD	AS	991750968

By signing this contract, we acknowledge having read the Sheridan Academic Integrity Policy

Responsibilities of the Project Leader include:

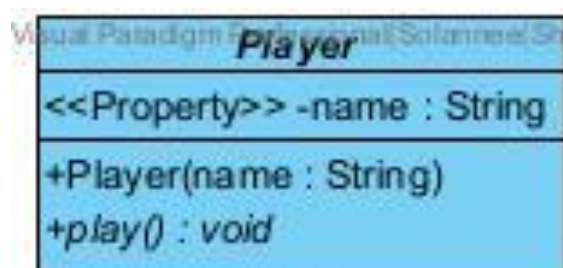
- Assigning tasks to other team members, including self, in a fair and equitable manner.
- Ensuring work is completed with accuracy, completeness and timeliness.
- Planning for task completion to ensure timelines are met.
- Notifying the professor of any issues in a timely manner so that corrective measures can be taken.
- Any other duties as deemed necessary for project completion.

What we will do if . . .

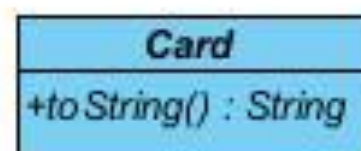
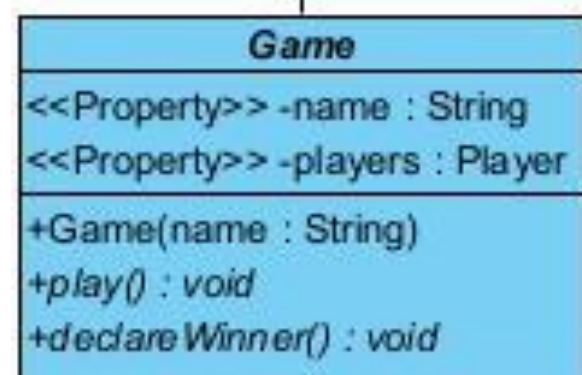
Scenario	Accepted initials	We agree to do the following (Put an X corresponding to your choice in each box)
Team member does not regularly attend team meetings and/or does not respond to communications in a timely manner.	PD VL NS AS	Project leader emails the student citing the concerns and cc's the professor so they are aware of the situation at the very onset <u> X </u> (Mandatory). a) <u> </u> In addition to above, the leader/team will (add your own content here):
Team member does not deliver component on time due to severe illness or extreme personal problem.	PD VL NS AS	a) Team absorbs workload temporarily <u> X </u> b) Team seeks advice from professor <u> </u> c) Team shifts target date if possible <u> </u> d) <u> </u> Other (specify):
Team member has difficulty delivering component on time due to lack of understanding or ability.	PD VL NS AS	a) Team reassigns component <u> </u> b) Team helps member <u> </u> c) Team member must ask professor for help <u> X </u> d) <u> </u> Other (specify):

Scenario	Accepted initials	We agree to do the following (Put an X corresponding to your choice in each box)
Team member does not deliver component on time due to lack of effort.	PD VL NS AS	a) Team absorbs workload <u> X </u> b) Team member(s) ask professor to request a Participation Form from <u>all</u> team members. This <i>may</i> result in individualized grades being awarded for a deliverable <u> </u> c) Both a. and b. above <u> </u> d) <u> </u> Other (specify):
Team cannot achieve consensus leaving one or more member(s) feeling that their voice(s) is/are not being heard in a decision which affects everyone.	PD VL NS AS	a) Team agrees to abide by majority vote <u> X </u> b) Team seeks advice from the professor <u> </u> c) <u> </u> Other (specify):
Team members do not share expectations for the quality of work on a particular deliverable.	PD VL NS AS	a) Team members will draw on each other's strengths to help bring the quality of the deliverable to a minimal acceptable level <u> X </u> b) Team votes on each submission's quality <u> </u> c) Team member(s) ask professor to request a Participation Form from all team members, which may result in individualized grades being awarded for a deliverable <u> </u> d) <u> </u> Other (specify):
Team member behaves in an unprofessional manner, e.g. being rude, uncooperative and/or making one or more	PD VL NS AS	a) Team agrees to avoid use of all vocabulary inappropriate to a business/college setting <u> X </u>

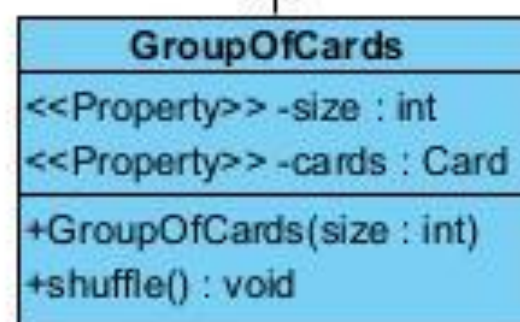
Scenario	Accepted initials	We agree to do the following (Put an X corresponding to your choice in each box)
member(s) feel uncomfortable.		b) Team attempts to resolve the issue by airing the problem at a team meeting ____ c) Team requests a meeting with the professor to discuss further ____ d) ____Other (specify):
There is a dominant team member who insists on making all decisions on the team's behalf leaving some team members feeling like subordinates rather than equal members	PD VL NS AS	a) Team will actively solicit consensus on all decisions which affect project direction by asking for each member's decision and vote ____ b) Team will express subordination feelings and attempt to resolve issue <u>X</u> c) Team seeks advice from the professor ____ d) ____Other (specify):
Team has a member who refuses to participate in decision making but complains to others that s/he wasn't consulted	PD VL NS AS	a) Team forces decision sharing by routinely voting on all issues ____ b) Team routinely checks with each other about perceived roles ____ c) Team discusses the matter at team meeting <u>X</u>



-players * (multiplicity at Player end)



-cards * (multiplicity at Card end)



SYST 17796 – DELIVERABLE 1

DESIGN DOCUMENT – WARGAME

1. Project Background and Description

The **War Card Game** is a competitive turn-based card game where players flip cards from their deck, and the player with the highest card wins the round. In the event of a tie, a "war" occurs where additional cards are drawn to determine the winner. The game continues until one player collects all the cards or a predetermined number of rounds is played.

The game is developed in **Java**, following **Object-Oriented Programming (OOP) principles** such as **encapsulation, inheritance, and modularity**. The current base code includes abstract classes (Card, Game, Player) and a concrete class (GroupOfCards) that provide a flexible foundation for implementing game rules. The design pattern used focuses on **encapsulation and delegation** for better code organization and maintainability.

2. Project Scope

Team Members and Roles

Name	Role
Pratham Doshi	Lead Developer
Vimal Lakhani	Tester / Developer
Azaan Shahzad	Project Designer
Neel Solanki	Documentation / Code Reviewer

Technical Scope

- The game will be a **console-based** War Card Game with future scope for graphical enhancements.
- The game will be considered complete when it meets the following conditions:
 - Implements full War game rules.
 - Supports 2, 4, and 6-player modes.
 - Displays player scores and winner determination logic.

3. High-Level Requirements

The new system must include the following:

- Ability for each player to **register and participate** in the game.
- Ability for the game to **determine and display the winner**.
- Ability for players to **track their score in real-time**.
- Ability to **handle tie situations (War rounds)**.
- A structured **turn-based game loop**.

4. Implementation Plan

Git Repository Information

- **Repository Name:** WarGame_Group7
- **GitHub Repository URL:** https://github.com/Coding-Pratham/WarGame_Group7.git
- **Usage:**
 - Each developer commits their work at the end of each day/week.
 - Feature branches are created for each major change.
 - Code reviews ensure quality and consistency.

Development Tools s Standards

- **IDE:** NetBeans
- **Version Control:** GitHub
- **UML Diagrams:** Visual Paradigm
- **Coding Standards:**
 - Consistent indentation and naming conventions.
 - Proper use of access modifiers for encapsulation.
 - Use of design patterns for modularity.

5. Design Considerations

➤ Encapsulation

- The Card class encapsulates **card value and suit** within private fields, ensuring data integrity.
- The Player class manages its hand of cards without exposing direct manipulation, allowing for controlled interactions.

➤ Delegation

- The Game class delegates **card management** to GroupOfCards, separating concerns and improving code organization.
- The Player class delegates **card drawing** to Deck, maintaining modular functionality.

➤ Flexibility s Maintainability

- The abstract Game class allows for future extensions, such as **multiplayer modes**.
- The modular design allows easy addition of **new game features** without rewriting core logic.